

INVESTIGATION OF MECHANISMS ASSOCIATED WITH EPOXY RESIN NANOCOMPOSITE DIELECTRICS

By

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A Thesis Submitted to the Graduate
Faculty of Rensselaer Polytechnic Institute
in Partial Fulfillment of the
Requirements for the Degree of
DOCTOR OF PHILOSOPHY

Major Subject: ELECTRIC POWER ENGINEERING

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December 2009

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ACKNOWLEDGMENT

The completion of this thesis owes to many people around me. First of all, I would like to express my sincere appreciation to my advisor, Professor Salon for his help and support for my Ph.D study. I am also very grateful for Professor Schadler's always-kind help and continuous support throughout the years of my Ph.D study.

I also thank Professor Leila Parsa and Professor Ishwara Bhat for their invaluable suggestion and help.

In addition, I would like to give my special thanks to Professor Nelson for his supervision and guidance from the very beginning of this research. During the years, his great insight on dielectrics helped me to grow as a scientist and his support and guidance makes him a great mentor.

During my thesis work, a lot of friends and colleagues helped me in various ways. I would like to thank Jittakorn Thiticharoenpong for his help on the initial training on the research topic. I would like to thank Dr. Dongling Ma, Dr. Robert Smith, Dr. Tong Liu, Dr. Xinxing Liu and Hui Le. I very much appreciate their enlightening discussions and information.

The financial support for my Ph.D study from my employer ISO New England is greatly appreciated.

I would like to thank the invaluable support of my family. My beloved father and mother have undergone all the hardships in bringing up and educating their daughter. They gave me the great love for all my life. My parents-in-law also gave me their encouragement and support for my study and life as much as they can.

A very special thank should go to my husband, Jian Mao, for his love, support and help ever since we met. Without him, this dream could never come true. His encouragement and companionship has turned my life into a great journey.

ABSTRACT

Epoxy resin has been widely used in the power industry for its superior insulating properties. By mixing micron scale fillers into epoxy resin, better thermal and mechanical properties can be achieved. However, in most cases, the introduction of the fillers will deteriorate epoxy resin's insulating properties. With the development of nanotechnology, the inclusion of nanometric fillers in polymers to form nanodielectrics has drawn much attention. the size of fillers reduces to nanometric level, the interface between the particles and polymer becomes the dominant factor in determining the composite's bulk properties.

This study examines the dielectric properties of epoxy resin based nanocomposites. Practical measurements show that nanocomposites have superior short-term breakdown strength and voltage endurance capability over microcomposites. The particles dispersion, the curing characteristics, and free volume of nanocomposites were first examined and compared to microcomposites. The results show that the nanoparticles could affect epoxy resin's crosslink density and increase the free volume of the composites. Dielectric spectroscopy measurements indicate that at elevated temperatures, nanocomposites have lower chain activation energy compared to microcomposites. Mechanistic study reveals that space charge plays a significant role in composite's properties. In nanocomposites the charge accumulation is reduced and the time constant of charge dissipation is much shorter. Furthermore, space charge in nanocomposites is featured a deeper trap depth of 1.57 eV compared to a trap depth of 1.21 eV in microcomposites. It was concluded that the interface between the epoxy resin matrix and the nanoparticles provides a higher conductivity path for the space charge to transport. The nanoparticles work as scattering centers and lead to the homocharge accumulation near the electrodes in nanocomposites. The improved breakdown strength and voltage endurance in nanocomposites is related to the shielding effect from the homocharge.

CHAPTER 1

BACKGROUND AND LITERATURE REVIEW

Epoxy resin has been widely used in the power industry since the 1950s. Because it is easily cast and has superior insulating properties and chemical and water resistance, epoxy resin has long been popular as an insulating material. Nowadays, epoxy resin is widely used in switch gear and cable termination insulation. It has also been used for rotating machine insulation by curing in situ [1] and in Gas Insulated Substation(GIS) as solid insulation [2].

As polymers were developed for use in high voltage insulation, additives/fillers were mixed with polymers to modify their mechanical and thermal-mechanical properties. In GIS, epoxy resin is usually filled with certain amount of micron-scale fillers, such as silica [3], alumina [4] or barium titanate [5], to achieve the same low thermal expansion as the metal conductors [2]. The conventional filled epoxy with low thermal expansion avoids exfoliating between the epoxy casting part and the conductor [2]. However, it has been observed that conventional fillers very often deteriorate the insulating properties of the epoxy resin. Usually with the addition of the conventional fillers, the dielectric loss of the epoxy resin will increase leading to higher permittivity [6, 7, 8, 9]. The Maxwell-Wagner interfacial polarization between the filler and polymer also causes space charge build up along the interface. Thus the aim for the new composite epoxy resin system is to maintain/improve the base resin's dielectric properties instead of deteriorating it, while at the same time preserve the thermal and mechanical advantage of the conventional composite system.

The application of nanotechnology in the semiconductor and biological fields has advanced significantly since the 1990s. Lewis [10] first predicted the future of nanotechnology in the field of dielectrics and insulation in 1994. He noted the importance of the interface at the nanometric scale. Frechette [11] also reviewed

the perspectives of nanodielectrics and their insulating properties in 2001. It took almost seven years for the first experimental data [13] to be published. In this paper, the unique characteristics of nanocomposites were demonstrated, such as lowered permittivity even with the addition of high permittivity particles, mitigated interfacial polarization characteristics. These initial striking findings cast light on the possible revolutionary impact of polymer nanocomposite dielectrics.

Nanodielectrics [11] consist of multi-component dielectrics possessing nanostructures, in presence of which resulting in the change of one or more of its dielectric properties. Frechette [11] divided nanodielectrics into two major categories: ceramics and polymer nanocomposite. Polymer nanocomposites are defined as polymers with nanometric fillers. The nanometric fillers are between 1 ~ 100 nm in size in at least one dimension.

Nanoparticles are intermediate in size between molecular and microscopic structures [14]. Thus their electronic and magnetic behavior are often dominated by quantum behavior [15]. This provides the potential for unique engineered materials [16] - [19].

Compared to conventional micron particles, a nano-sized particle contains fewer atoms and has a much higher percentage of atoms present at the particle boundaries [20]. One extreme example is, a metal nanocrystal of 1 nm in diameter, which will have nearly 100% of its atoms on its surface [21]. Since such a large fraction of the atoms reside on surfaces, when mixed with a polymer, this interface structure can play a significant role in affecting the properties of polymer composites [20].

Nanoscale particles also have much larger surface area compared to the same weight micron particles, which leads to a high surface-to-volume ratio. When the surface-to-volume ratio is high, the interface between the two phases becomes the dominant factor on the composite's dielectric properties [23].

Finally, nanoparticles have much higher chemical reactivity than conventional particles [26]. Surface treatment can be used to modify the interaction between the fillers and improve the dispersion of nanoparticles [24]. Moreover, nanoparticles

can be coated to form various bond with the polymer matrix for desired properties. For example, nanoparticles can be coated and initiate polymerization on their surface [25]. This provides possibilities for precise design of nanocomposite material properties. With surface modification of nanoparticles, certain properties can be significantly magnified while other undesired properties suppressed.

Their unique quantum access, large surface-area-to-volume ratio, and high surface chemical reactivity have drawn much research interest in this area. However, the fundamental mechanisms involved in nanocomposites is still under discussion. The focus of this study is to determine the effects of adding nanoparticles to epoxy resin matrix on the dielectric properties. The improvement of the dielectric properties, such as short-term breakdown strength, voltage endurance etc, are first presented. In order to obtain an understanding of the underlying mechanisms, the nanoparticle dispersion, the curing characteristics and free volume properties were examined and compared to conventional composites. Furthermore, the internal charge behavior was studied by pulsed electric-acoustic measurements, thermally stimulated current measurements and electroluminescence measurements. The polymer matrix dynamics with frequency and temperature were measured with broadband dielectric spectroscopy. Based on the experimental observations the underlying mechanisms are discussed.

In this chapter, the background of polymer nanocomposites and the current approaches of preparing nanocomposite are introduced in the first section. In the second section, current dielectric experimental results and mechanisms associated with polymer-inorganic filler nanocomposites are presented. In the end, the objectives of this work are addressed.

1.1 Epoxy Resin

Epoxy resin is a thermosetting epoxide polymer that polymerizes and crosslinks when mixed with a catalyzing agent or “hardener”. Depending on the chemical structure of the curing agent and curing conditions, epoxy resin can be categorized

into three different types [27]: phenolic glycidyl ethers, aromatic glycidyl amines, and cycloaliphatics. The most common used resin is diglycidyl ether of bisphenol-A(DGEBA).

The resin has to be cross-linked into a three-dimensional insoluble and infusible network. Two types of curing agents are widely used [28]: catalytic systems and polyfunctional cross-linking agents that link the epoxide resin molecules together (co-reactive systems). Some systems may involve both catalytic and condensation systems. For example, amine curing agents can cross-link epoxy resins either by a catalytic mechanism or by bridging across epoxy molecules. In general, primary and secondary amines act as co-reactive hardeners while tertiary amines are essentially catalytic. As a class, primary aliphatic amines form fast curing agents at room temperature. Primary and secondary aliphatic poly-amines allow good room temperature curing of DGEBA type of resins.

1.2 Polymer Nanocomposites

Polymer nanocomposites are defined as polymers with nanometric fillers. The nanometric fillers are 1-100nm in size [11]. Common polymer host matrices are LDPE, XLPE, epoxy resin, PMMA, etc. Common inorganic filler particle candidates include TiO₂, ZnO, SiO₂, etc. An ideal nanocomposite should be free of any agglomerates and contain optimum filler content [12].

There are various methods to mix the polymer and filler particles. For thermoplastic polymers, such as polyethylene, filler particles are typically mixed into the polymer during melt mixing. For thermoset polymers, particles are normally added before or during polymerization. However, it is often very difficult to disperse nanoparticles well in polymers especially at a commercial production scale. Nanometric filler particles with high surface energy tend to agglomerate. In addition, hydrophilic nanoparticles and hydrophobic polymers are not compatible in nature. All these could affect particle dispersion in the polymer matrix and thus result in poor interfacial interaction.

Different methods have been used to treat particles surface treatment to help the dispersion in the polymer matrix. For example, Gunko et al.[29] studied the features of fumed silica coverage with several kinds of silanes with both experimental and theoretical aspects. Depending on the pretreatment conditions, the attached silanes can reduce the hydrophilic properties of silica to different extents.

Grafting macromolecules onto inorganic particles is another popular method [30]. The polymer grafted particles can possess desired properties through a proper selection of species of graft polymers and the graft conditions so that the interfacial characteristics between the treated nanoparticles and the matrix can be tailor made when manufacturing nanocomposites. This can be done by 1) polymerizing monomers from active compounds covalently attached to the inorganic surface ('grafting from') or 2) polymers with reactive end groups react with the functional groups on the particles ('grafting to'). For example, Espiard and Guyot [31] treated silica nanoparticle surface with a silane coupling agent containing a double bond, i.e. methacryloxypropyltrimethoxysilane (MPTMS). Then, the modified silica was grafted with polymethylacrylate in an emulsion solution.

1.3 Experimental Results and Theories

1.3.1 Dielectric Breakdown Strength of Nanocomposites

Breakdown of dielectrics happens when a conducting path forms, which often involves the destruction of the polymer molecular structure. Under such circumstances the insulator will have been irreversibly converted to a conductive state.

Depending on the charge initiation and current multiplication process, dielectric breakdown can be categorized as intrinsic breakdown, electron avalanche breakdown, free-volume breakdown, thermal breakdown and electro-mechanical breakdown, etc [32]. In the free-volume theory, the breakdown is initiated by free electrons accelerated by an electric field in the largest hole present in the amorphous phase of the polymers [33]. In addition, breakdown can be caused by space charge and AC

treeing.

Zilg et al. [34] investigated the breakdown strength of two kinds of polymer nanocomposites containing nano-sized organophilic layered silicates–isotactic polypropylene, PP, and a copolymer ethylenevinylacetate, EVA. They found under the ramped-DC voltages, the addition of nano-sized silicates have an effect on the breakdown strength of the composites. In the case of EVA, the addition of nano silicates decreased the breakdown strength slightly. While in the case of PP, the breakdown strength increased with the addition of nano-silicate. On a Weibull plot, it was also found that the shape parameter value increases for nanocomposite, which indicates that the material becomes more homogeneous.

Hong et al. [35] mixed 49 nm ZnO particles into an LDPE matrix. The breakdown strength was measured and compared to the corresponding microcomposite. They found at relatively low particle loading, the nanocomposite and microcomposite had similar breakdown strength. At 40% loading by volume, the nanocomposite showed a significant margin over the microcomposite (Figure 1.1).

Sarathi et al. [36] studied the breakdown strength of epoxy resin/organophilic MMT (Montmorillonite) clay nanocomposite. The needle-plane samples were subjected to DC and AC fields respectively. Figure 1.2 shows the breakdown strength in relation to the particle loading level. The addition of nanometric MMT increased the breakdown strength compared to the base resin under both DC and AC conditions. The 5 wt% loading level was found to be optimal here. The nanocomposite showed higher breakdown strength under positive DC field.

Cao et al. [37] studied a series of polyamide based nanocomposites. Different kinds of nano-sized particles at different loading levels were added to polyamide and the breakdown strength was measured. Figure 1.3 shows the 2-parameter Weibull statistic plot of alumina/polyamide nanocomposite breakdown strength under different loading levels. The electrical breakdown strength of polyimide(PI) is not affected by the addition of alumina nanofillers up to 10 wt% loadings.

Treeing experiments are usually helpful in the investigation of breakdown mech-